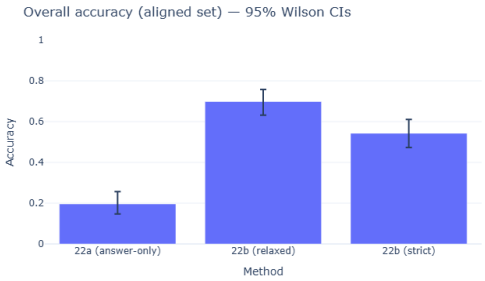
Method	Accuracy (%)	95% CI	Δ vs. Baseline (%)
Answer-only (k=3)	19.6	[14.7, 25.7]	—
PC-CoT (Relaxed)	69.8	[63.1, 75.8]	+50.3***
PC-CoT (Strict)	54.3	[47.3, 61.0]	+34.7***

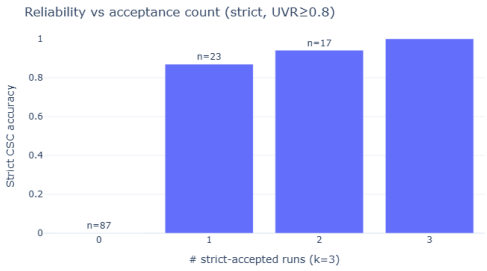
Table 1: Overall accuracy on aligned GSM8K subset ($n = 199$). Numbers are proportions expressed as percentages. Confidence intervals are at 95%. Significance: $^{***}p < 10^{-14}$ (two-proportion z-test).

the PC-CoT method at around 200 examples (which was more cost effective against the GPT5 API). All experiments used $k = 3$ samples per question with identical token budgets across methods.

PC-CoT demonstrates significant improvements over the answer-only baseline. As shown in Table 1 and Figure 1, typed certification yields gains in both overall accuracy and reliability. The relaxed gate achieves a 50.3 percentage point gain ($z = 11.68$, $p \approx 0$), while the strict gate maintains a 34.7 point advantage ($z = 7.68$, $p = 1.6 \times 10^{-14}$). These gains arise from typed filtering converting low-precision chains into high-precision candidates before voting.

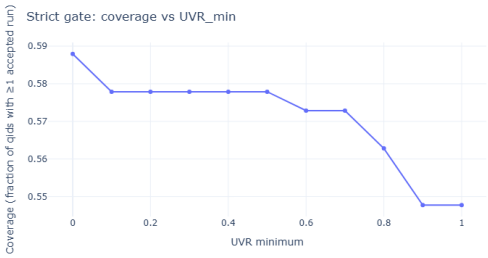


(a) Overall accuracy (95% Wilson CI)

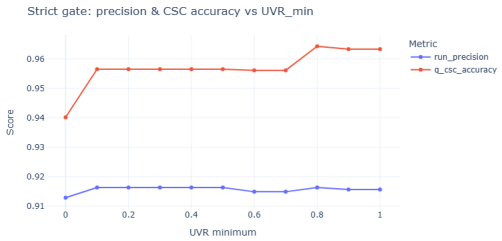


(b) Reliability vs. acceptance count

Figure 1: Overall accuracy compared to the answer-only baseline (left) and reliability of strict CSC predictions as a function of the number of accepted runs (right). These plots jointly illustrate that accuracy improves sharply with typed certification, and that reliability continues to increase when multiple certified runs agree.



(a) Coverage vs. UVR minimum



(b) Precision & CSC accuracy vs. UVR minimum

Figure 2: Strict gate sensitivity analysis. Coverage declines as UVR thresholds tighten (left), while precision and CSC accuracy improve (right), highlighting the tradeoff between coverage and selectivity. This demonstrates that unit consistency checks are a useful control knob: higher thresholds reduce spurious acceptance but at the cost of lower problem coverage.

5.2 Certification Selectivity and Precision

Certification acts as a significant precision filter, transforming noisy natural language reasoning chains into verifiable typed programs. As shown in Table 2, accepted runs under the relaxed gate

Metric	Accepted Runs	Rejected Runs	Precision Gain (%)
Relaxed Gate (EVR $\geq$ 0.30, PE required)			
Run-level accuracy	87.2% (511/586)	40.9% (251/614)	+46.3
Questions with $\geq$1 certified	70.4% (140/199)	—	—
Strict Gate (EVR $\geq$ 0.80, PE required, Consistency, UVR $\geq$ 0.80)			
Run-level accuracy	91.6% (471/514)	42.4% (291/686)	+49.2
Questions with $\geq$1 certified	56.3% (112/199)	—	—

Table 2: Certification selectivity analysis on GSM8K ($n = 199$). Accepted runs under both gates achieve dramatically higher accuracy than rejected runs, yielding nearly 50 percentage point precision gains from certification.

reach 87.2% accuracy compared with only 40.9% for rejected runs. The stricter gate pushes precision even higher (91.6%), while rejected runs remain at baseline levels near 42%. This nearly 50 point precision gap demonstrates that type-based certification reliably separates well-formed reasoning traces from ill-typed or incoherent ones.

Coverage at the question level shows the expected trade-off: the relaxed gate certifies at least one run for 70.4% of questions, while the strict gate certifies fewer (56.3%) but with higher precision. Figure 2 demonstrates that tightening the UVR threshold increases precision at the expense of coverage, underscoring that certification acts as a tunable control knob. This tunability allows PC-CoT to adapt to different application settings—for instance, using relaxed gates for exploratory reasoning where recall is important, and strict gates for safety-critical contexts where dimensional validity and consistency must be guaranteed.

5.3 Error Decomposition and Coverage Analysis

Decomposing performance on the aligned set reveals that PC-CoT solves far more problems uniquely than the baseline. Under the relaxed gate it contributes 104 unique wins, and under the strict gate 79 unique wins, compared with only 4 and 10 questions, respectively, that are solved exclusively by the answer-only baseline. At the same time, there remains a coverage gap: 87 questions do not have any strict-certified run, which helps explain the difference in performance between the relaxed and strict gates. PC-CoT uniquely solves 10-25× more problems. This demonstrates that typed certification fundamentally affects the reasoning landscape rather than merely filtering existing capabilities.

5.4 Computational Faithfulness: Program-CoT Alignment

We validated faithfulness by aligning generated programs with natural language CoT (Turpin et al., 2023a). High alignment rates in Table 3 support our central hypothesis: typed programs capture computational structure rather than post-hoc rationalisations. Moreover, the fact that nearly one-fifth of runs show low alignment underscores that certification is selective—faithful reasoning emerges in structured cases, while ill-typed or weakly aligned traces are systematically filtered out. This does not mean that the particular reasoning traces used in the experiments are necessarily causally driving the underlying model per se, but it does speak to the utility of the CHC method of (extracting programmatic representation from reasoning traces) which can in principle be adapted to more mechanistic data (e.g. activation structure) in reasoning models.

5.5 Ablation Studies

To assess the contribution of each component of PC-CoT, we ran ablation experiments in which we systematically removed type checking, path requirements, SCS, or soft decoding constraints, and compared the resulting accuracies to the full model. As Table 4 shows, each component contributes substantially to overall performance. CSC provides the largest gain (+34.1%), while soft constraints outperform hard grammar enforcement by 20.9%, validating our design choices. Taken together, the ablations show that typed verification, proof-path checking, and selective consistency are useful components in operationalising the CHC for analysis of LLM reasoning.

Alignment Level	Frequency	Example
Full alignment (100%)	43%	Every operation traceable to CoT line
Partial alignment (50-99%)	38%	Most operations match, minor paraphrasing
Low alignment (<50%)	19%	Significant structural differences

Table 3: Alignment between typed programs and natural language CoT traces. In total, 81% of certified runs show substantial alignment (full or partial), indicating that most certified outputs preserve a coherent mapping between program steps and CoT reasoning.

Configuration	Accuracy	Δ vs. Full
Full PC-CoT (Relaxed)	69.8%	—
Without type checking	41.2%	-28.6%
Without path requirement	52.3%	-17.5%
Without CSC (use all runs)	35.7%	-34.1%
Hard constraints (L4)	48.9%	-20.9%

Table 4: Ablation study on GSM8K ($n = 199$). Each row removes one component of PC-CoT: type checking, path requirement, CSC (all runs aggregated without certification), or soft constraints. Removing certification or using rigid grammar significantly degrades performance.

6 Discussion

Our results show cases in which PC-CoT operationalises a key prospective application of the CHC to language reasoning models: faithful reasoning explanations should correspond to well-typed programs that compute their conclusions. Our findings indicate the existence of a typed reasoning gradient: a strong correlation between the degree of type-checkable structure in a CoT trace and its final correctness. This suggests that faithfulness is not a binary property but a spectrum, with important implications for interpretability and reliability. By applying the CHC correspondence during generation rather than post-hoc, we can in certain cases transform noisy reasoning chains into proof-carrying artefacts with formally verifiable properties. Our results also reveal that PC-CoT reasoning exhibits a *typed reasoning gradient*: while not all CoT admits complete typed proofs, those that do achieve dramatically higher accuracy (see the Appendix). This gradient suggests that typed structure is not binary but exists on a spectrum, with important implications for interpretability and reliability.

6.1 Implications

PC-CoT provides a method to approximate constructive proof—typed programs from reasoning traces, which enhances interpretability. This has several advantages: (a) *Verifiability*: Typed programs can be independently executed and checked; (b) *Compositionality*: Complex reasoning decomposes into typed sub-proofs; (c) *Debugging*: Failed type checks pinpoint reasoning errors. Our results provide empirical support for viewing LLM reasoning through the lens of type theory. The strong correlation between type-checking success and correctness (91.6% precision for strict-certified runs) suggests that successful reasoning inherently exhibits program-like structure, even when expressed in natural language. This aligns with mechanistic interpretability findings that CoT induces modular internal computation (Chen et al., 2025a), but goes further by providing an external, verifiable signature of faithful reasoning. The 50+ percentage point accuracy gains demonstrate that typed certification may have practical benefits for certification-critical activities identified in the literature (Shah et al., 2025; Benton et al., 2024; METR, 2024; Chan et al., 2025).

6.2 Limitations and Future Work

However, our results also highlight limitations. The 40-60% ceiling on complete typing suggests that much LLM reasoning involves implicit steps, commonsense jumps, or genuinely non-compositional computation that resists formal typing. Other limitations are worth noting: (a) *Domain specificity*: Our type system targets arithmetic reasoning; extending to abstract reasoning requires richer type theories; (b) *Soft vs hard typing*: We filtered at selection time rather than enforcing hard constraints